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***B.Tech. Degree III Semester Supplementary Examination
April 2018***

**CS 15 - 1305 PRINCIPLES OF PROGRAMMING LANGUAGES
(2015 Scheme)**

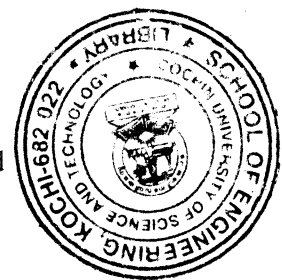
Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Write notes on Language Evaluation criteria.
- (b) What are the factors affecting the readability of programs?
- (c) Define static binding; and dynamic binding.
- (d) What are the l-value and r-value of a variable?
- (e) Briefly mention the features of Smalltalk.
- (f) With a suitable example, briefly explain data abstraction and encapsulation.
- (g) Can you define a reference variable for an abstract class? What use would such a variable have?
- (h) Write a note on horn clauses.
- (i) Why are CAR and CDR so named?
- (j) Explain the applications of functional language.



PART B

(4 × 10 = 40)

- II. What are the formal methods of describing syntax and semantics? (10)
- OR**
- III. Explain Imperative, Functional, Object Oriented and Logic programming with suitable examples. (10)
- IV. (a) Describe the scope and life time of a variable. (7)
- (b) Briefly explain the named constants. (3)
- OR**
- V. Describe with suitable examples the pass-by-value, pass-by-result, pass-by-value result, and pass-by-reference parameter-passing methods. (10)
- VI. Discuss the design and implementation issues associated with programming languages. (10)
- OR**
- VII. With suitable examples, explain polymorphism and inheritance. (10)
- VIII. Briefly describe the features of functional language LISP. Also explain the lambda calculus. (10)
- OR**
- IX. Explain the logic programming language PROLOG with suitable examples. (10)